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Management Conception Challenges in the Post-Covid-19 Era and the Stra.Tech.Man Analysis

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Abstract: The management of organisations is a continuously evolving and transforming science, which first emerged as a theoretical instrument and method to improve efficiency and effectiveness. This chapter explores the conceptual evolution of management theory from an elliptical and critical perspective. It examines

the relevant scholarly management literature and is oriented towards understanding and synthesising different conceptual trends in change management. This conceptual paper concludes that a refocused theoretical framework oriented towards change management in 'Stra.Tech.Man' terms (strategy-technology-management synthesis) is necessary for organisational innovation and effective adaptation to the newly emerging conditions.

Keywords: management evolution; change management; Stra.Tech.Man approach; post-COVID-19 era; innovation; evolutionary management

1. Introduction

Nowadays, global capitalism seems to be entering a new phase, creating a significantly differentiated field of action for all socioeconomic organisations. This emerging phase of the global economy—characterised by the crisis and restructuring of globalisation and the fourth industrial revolution in the post-COVID-19 era—appears to accelerate pre-existing trends (Bragazzi, 2020; Schwab, 2016; Vlados et al., 2018). According to a neo-Schumpeterian interpretation (Freeman, 2009; Schumpeter, 1939), in today's subversive circumstances, it is expected that various ageing industries will be driven at an accelerated rate to decay and extinction. New, more dynamic, and more effective organisational systems (firms and sectors of economic activity) will take their place. This innovation stream, which forms the incubation basis for 'creative destruction' (in the Schumpeterian approach), increasingly seems to be the result of the gradual 'coupling' of cyber-physical systems, where the distinction between physical and digital dimensions becomes progressively indistinguishable (Frazzon et al., 2013; Schlick, 2012). In this context, digital transformation and innovation (through a refocused change management framework) become urgent—simultaneously at the individual, firm, local, regional, national, and global level (Clegg et al., 2021; Hanelt et al., 2020).

What does management mean, and what must managers do in today's circumstances? A thorough understanding requires examining the variety and synthesis of different approaches rooted in the foundations of management. H. Fayol's (1916) classical definition of management seems to be still relevant and appropriate in its clarity. It is argued that all managers perform specific activities that concern forecasting and planning, organising, commanding, coordinating, and controlling. This fundamental definition of management is still valid even after approximately one hundred years of scientific evolution in the field (Baran et al., 2019; Palomino et al., 2011). It is argued that these dimensions will be significantly restructured in the emerging altering global conditions (Anker, 2021; Gould et al., 2017; Van Hoek et al., 2020).

A wide variety of comprehensive definitions of the systematic management process is available nowadays, dating back to the second half of the twentieth century, where management texts began to proliferate. From an introductory and all-inclusive definition, it could be said that managing means doing things through other people, being the coordinated way of achieving individual and organisational objectives. According to Koontz and O'Donnell (1955, p. 64), '*Since enterprise management is concerned with achieving an objective or objectives through the coordinated effort of a group of persons, it is clear that groups must be led and that it is the manager who must lead*.' Similarly, Brech (1975, p. 19) has argued that '*Management is a social process entailing responsibility for the effective and economical planning and regulation of the operations of an enterprise, in fulfilment of given purposes or tasks*.' Therefore, in the context of the relevant scientific literature, it has been clear that, in turbulent conditions, responsiveness to customers, constant innovation,

organisation-wide partnership, and proactive, change-oriented leadership are necessary (Kim & Mauborgne, 2015; Laudicina & Peterson, 2016).

However, to what extent do current management approaches need to be refocused and re-enriched in the light of the new and profoundly subversive global developments caused by COVID-19? This question drives us to explore how the science of management has been conceptually evolving. This chapter investigates crucial historical configurations of management from a critical and elliptical perspective, concluding on a new change management approach and points for further enrichment. For this framework of understanding, a critical examination of the management literature is attempted, oriented towards studying conceptual trends to build a new and restructured approach (Snyder, 2019; Wee & Banister, 2016).

The following section explores some aspects of the fundamental schools of management, observed from the perspective of change and evolution. It first analyses the early 'classical management' developments from the late-nineteenth and early-twentieth-century and second the late twentieth century's Japanese management philosophy and practice. The subsequent section focuses on various recalibrations set forth within the dynamics of contemporary and flexible management approaches. Finally, the concluding section suggests a restructured theoretical scheme, emphasising how organisations can innovate by effectively synthesising the spheres of strategy, technology, and management (the Stra.Tech.Man approach), creating the necessary and corresponding change management mechanisms in today's turbulent times.

2. The central 'conceptual traditions' in management and the underlying problematics of change

Various scholars keep posing the following question: 'What is management? Art or science?' (Bruch & Ghoshal, 2004; Meisiek & Barry, 2014). It could be said that it is primarily a practice that aims to make social organisations more effective in achieving their goals. However, from the 'art' perspective, managers look to find the solution directly against a problem, based on their practical experience and 'intuition.' Then, they proceed to the next issue since the old problem has no 'direct' connection with the new one. From one crisis to another, they feel that their intuition sharpens and makes their managerial instinct more reliable. For them, usually, management is just a matter of talent, inspiration, and nature, and, according to this rationale, the ability to manage cannot be taught. In a similar vein, for the 'practitioners of management,' there is only management's daily practice. However, they are not right. Today, in the day-to-day operational practice, mostly the opposite seems to happen (Aucoin, 2018; Custovic & Insaurralde, 2015).

For others, management is only a sum of unchangeable, abstract, and ultimately inapplicable 'theoretical' principles (Cole, 2004; Tripathi & Reddy, 2012). For them, management is a 'fossilised' and dogmatic area, tending to believe that what was learned several years ago in student life will always remain valid and effective in operational practice. The proponents of this 'pseudo-scientific' approach, deliberately or not, indirectly justify the older and 'practically experienced' managers since old management theories never turn obsolete (Barr & Ben Shaḥar, 2019; Gat-Tilman et al., 2007).

From a synthesising perspective, the scientific approach to management can never be cut off from the ever-transforming empirical reality since it always begins with observed facts. It has to synthesise at the theory

level, predict, and control its predictions' accuracy by returning to experienced observations. This fact is valid in the methodological cycle of all empirical sciences (Figure 1).

[Figure 1 about here]

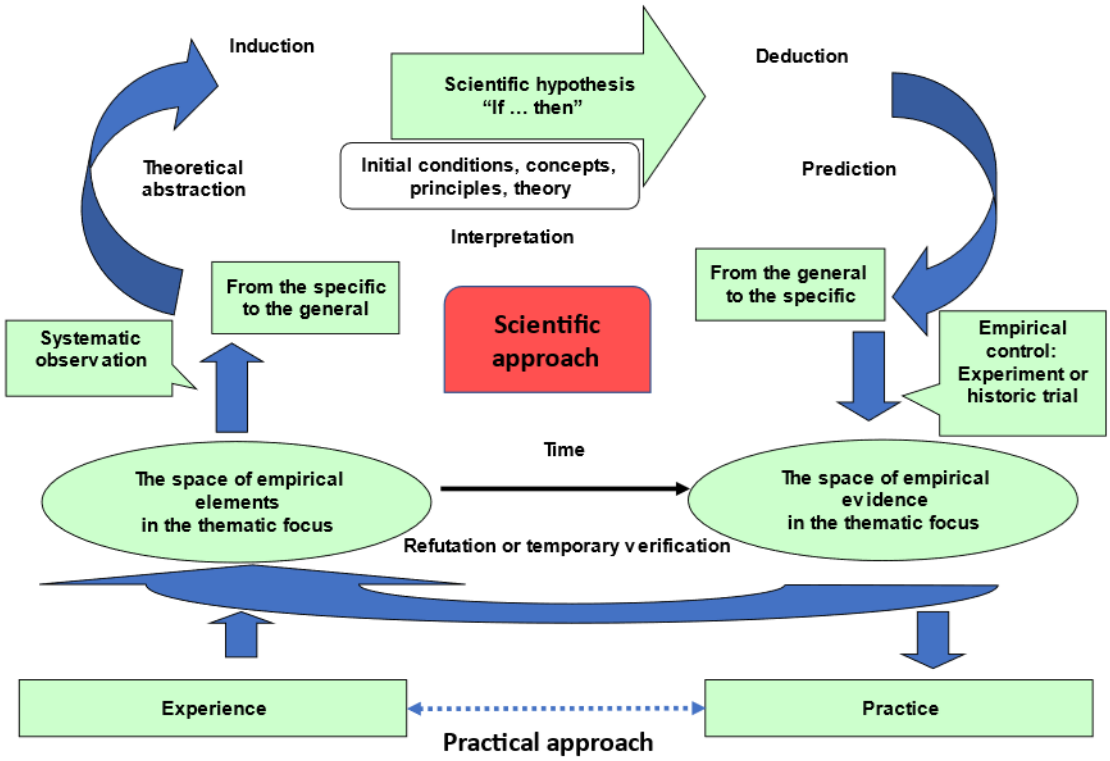


Figure 1. The fundamental methodological cycle of empirical sciences in a Popperian orientation (Vlados, 2020)

The practical approach arising from empirical reality must be different from the scientific process (Losee, 1972; Popper, 1934, 1963, 1966; Schick, 2000). The scientific method begins with experienced findings classified in the various thematic fields. This scientific approach suggests a theoretical abstraction from the 'specific' to the 'general' (induction) and builds an 'if-then' hypothesis articulated in derived concepts, principles, and theories. In the deduction step, the successful methodological cycle means accurate predictions by returning to the 'specific.' The researcher must empirically control the interpretation and forecasts at this stage. Finally, the theoretical hypothesis is confirmed or not by the facts, being subject to 'rejection' by a new scientific synthesis after being used in practical application.

Therefore, each methodological cycle follows an interaction between theory and experience-based practice and, thus, the purpose of theory is to embrace the practice itself as entirely as possible. Experience can only be the source, the cradle of each theoretical proposition, and at the same time, the necessary field of control. Experienced observations can only be the fundamental starting point and, at the same time, the theory's conclusion. In the end, science is not realised through 'big words' or appeals to some 'authorities.' Implementing this continuous cyclical relationship of 'practice-theory-practice' creates science (Derrida, 2019; Kuhn, 1962;

Lakatos, 1968).

Enlightening towards this direction are Drucker's (2002) notes on management's basic assumptions, arguing that this is a social science, predominantly expressed as handling business activity in a relatively optimal organisational structure and way to manage people. Drucker also stresses that technologies, markets, and end-uses are usually statically considered since management is only internally focused.

Overall, could the science of management ever stop evolving? Management is not a static field of expertise, but on the contrary, it attracts attention from multiple and interdisciplinary theoretical domains (Cezarino et al., 2016; Hacklin & Wallin, 2013). Management and organising open the path for new, fertile scientific directions, theoretical and applied simultaneously, heading towards profound renewal and rapid methodological developments. In this sense, a multi-collective and open management approach based on the different perspectives seems imperative for gaining valuable insights on this field (Cummings & Bridgman, 2019; Witzel, 2017).

2.1. Classical foundations of management and their limitations

Management as an empirical science has been developing for over one hundred years. With the advent of the eighteenth century's industrial revolution, the first studies on the science of management began to take shape. The first authors, such as Babbage (1835), realised the need for systematically studying work management and standardising business operations. With the appearance of mass production, this need became more intense. H. Towne (1886) formulated the first such scientific approach by presenting the work entitled 'The engineer as an economist' at the American Society of Mechanical Engineers. Towne brought forward the need for engineers to study management scientifically.

Taylor and other engineers across the USA and Europe began to produce relevant management questions and perspectives. They implicitly or explicitly suggested that the organisation resembles a 'machine' and the worker is the replaceable cogwheel (Emiliani, 2018; Jürgens et al., 1993). The first scholars of this origin suggested strict ways to handle organisational activities, usually applying inflexible hierarchical methods and an authoritarian style of corporate governance (Mahmood et al., 2012). Later, H. Fayol (1916) incorporated the seminal elements of management skills that can be learned and acquired through training, based on the fundamental functions of planning, organising, commanding, and controlling.

M. Weber's bureaucratic model is also a significant development, arguing that bureaucracy is the most effective way to manage large organisations and preserve order, optimise efficacy, and eradicate favouritism (Weber, 1947, 1949).[1] However, this bureaucracy also can threaten individual freedom, trapping individuals in an impersonal 'iron cage' of rule-based, rational control. Weber also observed at least six points to constitute this formal and bureaucratic structure in the organisation. The division of labour favours specialisation, with jobs broken down into simple routines and well-defined tasks. The bureaucracy should also have an authority hierarchy, meaning a hierarchically organised and transparent chain of command. Formal selection is another attribute, referring to people chosen for jobs based on their specific technical qualifications. Official rules and regulations are predetermined written rules and standard operating processes. Impersonality is also promoted since a uniform application of protocols and controls occurs, not according to specific personali-

[1] Weber wrote all his books in German. Original titles published after his death (1920) are compilations of unfinished works.

ties. Weber suggested career orientation as the last attribute since managers are career professionals and not owners of the units they manage.

Taylor is the founder of scientific management in the USA, based on the following triple functions and research: study of work, division of this work into smaller parts, and defining the most efficient way to execute them (Taylor, 1911). Taylor's approach to understanding work and improving the employee's efficiency is known as the search for the scientifically defined 'one best way.' From this perspective, scientific management is defined as a systematic method of determining the best way to do a job and specifying the skills needed to perform it (Khurana, 2009). This work's originality consisted of the fact that when tradition and inspiration ruled, Taylor managed to apply the analytical method of the logical structure of labour, which engineers mainly perform.

In terms of principles, at least four are identified. First, any work's scientific study must be done by a team of experts and not by the production staff. Timing and scientific observation (based on 'narrow' quantitative measurement) introduced a 'one best way' to get the job done, strictly defining the functional procedures and the amount of work an employee has to provide per post. Second, any worker must be selected, trained, and developed according to the science of management. The third principle (and combination of the previous two) makes science applicable to the workers, although the expert is still responsible for scientific observation and conception while the worker executes. At this point, Taylor argues, many practical managers fail since they do not want to change their methods in the light of scientific information. Fourth, there is an almost equal division of labour and responsibility between managers and workers. Managers are responsible for the tasks that can do better than workers.

Therefore, within the mass corporation, Taylorism promotes the progressive deepening of labour division, the intensive mechanisation of the production process, the multi-level administrative hierarchy, and the complete separation between the conception (design) and execution of projects (D. Nelson, 1992). Gilbreths are also pioneers in using time and motion studies to improve employee efficiency. F. Gilbreth (1912) invented a system of sorting fifteen movements in work and analysing the workers' actions, called work simplification. Taylor's associate, H. Gantt (1901)**,** dealt more specifically with and investigated the use of time-scheduling methods, creating a chart that depicts a work's timetable and proposing a worker's productivity task and bonus system. At approximately the same period, H. Ford promoted three fundamental innovations in the homonymous automobile factory that later led to a whole new management conception—Fordism (Raushenbush, 1937). First, work analysis is required to achieve the highest possible efficiency. Second, installing easy-to-use machinery is imperative since most workers, who used to be farmers, have no knowledge or training or education. Third, Ford introduced the assembly line, one of the most significant innovations in manufacturing.

Overall, the classical tradition in management is fully integrated into the dominant conception of Taylor. All administrative phenomena are mechanistically conceptualised in the different socioeconomic organisations that embrace the classical management rationale. The division of labour, the multi-layered and strict formality, the absolute centralisation in decision-making, and the 'bottom's sterilisation' against any initiative and participation has worked sufficiently well and effectively in the Western industrial world—and in the USSR experiment in the corresponding form of Stakhanovism (Bedeian & Phillips, 1990; Peaucelle, 2000). In conditions of relative stability and little uncertainty, it is a fact that this integrated classical management logic has

various virtues—systematic cost reduction, increasing scale economies, effective use of poorly trained human resources. However, what happens when external conditions cease to be smooth, easily prescribed and linearly evolving? This is where classical management's practical problems appear (Belasco, 1991; Spender, 1996).

2.2. The foundations of the critique towards the classical management perspective

The behavioural school suggests that organisations must not deny the human and social dimensions in work (Sorge & Warner, 1997). More specifically, E. Mayo (1930, 1933) was the founder of the human relations movement in management, arguing that experts trying to increase employee efficiency should also consider the human factor. The Hawthorne experiments conducted by Mayo also revealed that work productivity is related to social and psychological traits, as much as to work itself (Institute of Management Foundation, 1998). Mayo and his colleagues turned the managers' attention to employees' feelings for improving productivity. In this context, an effective manager needs to understand management techniques and consider all those factors that improve human relationships within a business unit.

From a similar research orientation, M. Follett (1941) suggested the cooperation method to solve problems at work, arguing that there should not be over-supervision of the regular workers. The principle of 'integration' was advocated, meaning a non-coercive power-sharing based on 'power with' instead of 'power over.' Also, Ch. Barnard (1948, 1949) introduced the theory of authority, in which it is postulated that employees understand whether a superior's order is legal and acceptable and when it is not. Personal goals and interests determine the acceptance and execution of commands, while management's effectiveness depends on the employee's welcoming of legality. Therefore, Barnard tried to prove that only orders considered acceptable are most likely to be executed based on the light of the workers' goals and personal interests. In the same behavioural stream of thought, D. McGregor's (1966) work is also significant. McGregor showed how employees become motivated by authoritative direction and control or integration and self-control, called theory X and theory Y, respectively. McGregor broke down earlier management styles, establishing the Y theory as a humane management approach. This approach would lead to the introduction of work plans that could spur employees' interest, promote individual development, improve the working environment, and increase productivity (Sorensen & Minahan, 2011).

In this conceptual evolution of management, another theoretical tradition was born and further investigated under the title 'operational research,' in which performance measurement is the primary method (Beer, 1968). H. Simon (1947, 1955), known for researching decision-making and information processes, found that business executives have no direct access to perfect information and can never take such flawless decisions to improve their results. Business administration can act based only on the first information on the problem. This decision rationale seeks the 'satisficing solution' instead of anything optimal. Simon replaced the rigid 'homo economicus' model with the 'bounded rationality' decision-making framework (Kalantari, 2010).

The contingency approach also critiques the mainstream management theory (Ghofar, 2016; Itzkowitz, 1996). This theoretical framework suggests that the practical solution to managerial problems depends on the unique and specific conditions of the issue at hand, observed from an eclectic rather than a dogmatic perspective. According to the contingency approach, no general solution to a management problem exists, but on the contrary, specific needs depending on the situation must be considered. Contingency theorists argue that management

problem-solving requires the synthesis of different perspectives, initiating from situation analysis and ending up with the formulation, evaluation, and suggestion of possible solutions (Battilana & Casciaro, 2012; Hanisch & Wald, 2012).

Most significantly, an approach that functioned as a comprehensive theoretical repositioning was total quality management (TQM)—or 'Japanese management' from a broader perspective. In this management orientation, the motto is 'quality first' and 'understand the customer,' both internal and external, systematically participate, improve, and continuously innovate (Deming, 1986; Juran, 1964). Total quality is achieved with all the active commitment and involvement since it is a continuous process focused on quality dimensions. Based on TQM's theory and practice, scholars have suggested various ways to simplify the implementation (Doney, 2019; Sharma, 2018).

An approach based on Japanese management principles is 'Theory Z'—a counterproposal to McGregor's X and Y theory. This management stream suggests that teams rather than individual managers take the most effective decisions (Ouchi, 1981). Theory Z supports consensual decision-making since the group has access and can filter more data, thus leading to better decisions. In this management direction, quality in working life is the primary concern, advocating and supporting life-long employment and participative management as a 'downward power': the employees' opinions are considered in decision-making. Overall, the basic principles of Japanese management philosophy, which were significantly different from Western forms of planning and organising during the early days of globalisation (the 1980s), were based on various participative principles (Jackson & Tomioka, 2004; Nonaka & Takeuchi, 1995; Stewart, 1996).

In sum, this critique on the classical management principles—that has been developing at least since the second half of the twentieth century—has highlighted entirely new central dimensions in management theory. These new management conceptualisations are pivotal for the contemporary management of all kinds and sizes of flexible organisations (Nandakumar et al., 2014; Vecchiato, 2015). From this critical perspective, against classical management, the emphasis is attributed to workers' human side, who have specific wants and continuously develop their personality and intellect to improve organisational performance. Significance is attached to group decision-making and participation at all hierarchical levels, systematic enrichment of jobs, and constant reinforcement of people. This focus on continually improving how people and groups adapt to fast-paced and transforming environments offers high-quality standards and undiminished commitment to innovativeness and flexibility (Anning-Dorson & Nyamekye, 2020; Bianchi & Labory, 2012; Dibrell et al., 2014).

3. Towards the post-COVID-19 management era

This section focuses on understanding the current challenges and prospects for management methods and definitions. Contemporary management refers to how individuals' achievements and goals can be achieved as constituents of the organisational goals (Certo & Certo, 2019; Spender, 2015). In this respect, today's management conceives each socioeconomic organisation as dialectically co-determined with its external environment, exceeding any narrow specialisation of the day-to-day experience. This knowledge-aligned management approach also strives to adapt to the rapidly transforming and restructured global environment (Owen, 2011).

Therefore, there can never be a static definition for the science of management, especially in today's conditions emerging in the post-COVID-19 era. However, there is not (and there will never be) a single and 'right' definition of management, accepted by all field experts since everything moves forward and evolves. Looking for a unique and 'proper' management definition (as well as a definition of any other scientific subject) can be anti-scientific (Holton, 1997; Levins, 1996). This would mean that the science of management is doomed to permanent 'sterility' and cognitive rigidity.

It should be mentioned that, towards offering a synthesising and evolutionary definition of management, all alternative definitions are mainly convergent in their procedural focus. Most contemporary managerial perspectives conceptualise management as a process that allows organisations to define and achieve objectives through comprehensible planning, effective organising, reliable staffing, and aptly controlling their available resources—in this process, coordinating acts as the force that binds together the other four functions (Andrikopoulos, 2009; Bateman et al., 2016; Williams, 2017). Concerning staffing, particular interest is placed in offering a framework for employees' strong commitment by motivating them. However, less research is underway on how management relates as an internal organisational process with the external global environment.

In this research direction, the significance is attributed to how management theory can cross the boundaries of 'mechanics' towards an organic conception of socioeconomic dynamics. Modern management theorists differentiate themselves from the analytical foundations of Taylorism, conceptualising the firm as an evolving and adaptive 'organism' instead of a passive 'machine' (Farjoun, 2002; Kessler et al., 2017). The 'living organisation' concept finds explanatory interest in perspectives calling for a clear theoretical transition from engineering to 'business biology' (Geus, 2002; Vlados, 2019c; Wagner, 2007). According to Battram (1999), in the new biological paradigm for business, the organisation should be treated as a 'garden' instead of a 'factory' while staff development resembles 'plant treatment' instead of 'annual maintenance.' In this context, the organisation does not need occasional oversight but continuous interaction among all organisational components because it is a complex and adaptive system, open to external influences, which can either grow or decline—never stays the same. In this 'ecosystemic' approach, improvement comes from shared values instead of extensively detailed procedures (Iansiti & Levien, 2004; Moore, 2015). More comprehensively, these 'biological' analyses of the economy show the gradual transition from a mechanistic to an evolutionary perspective (Figure 2).

[Figure 2 about here]

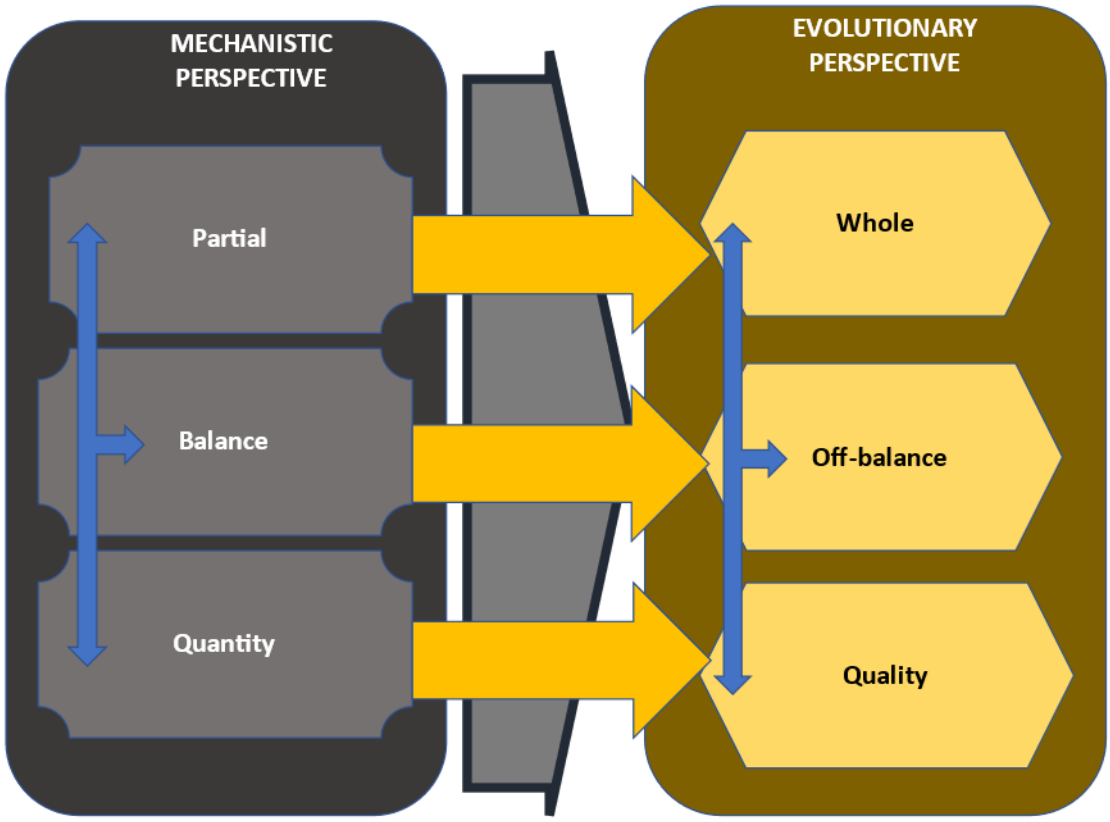


Figure 2. From a mechanistic to an organic and evolutionary perspective (Vlados et al., 2018)

This 'organic' analysis helps us realise that socioeconomic dynamics must be treated from a unifying perspective, focused on the continuous imbalances exerted upon all systems instead of static equilibria. Moreover, any over-concentration to quantitative accumulations and variances must be replaced by the profound qualitative transformations occurring under the surface. This new organic perspective also influences all contemporary trends in the evolutionary conception of management.

More specifically, evolutionary management suggests the always-temporary organisational structure and procedures. According to Endres (2018), no general principles or specific details of evolutionary management exist because it only refers to a framework that dictates how the structure, relationships, or systems schematise themselves. Therefore, evolutionary management concerns finding the proper behaviours and organisational routines for flexibility and adaptability; these two factors also are critical abilities for strengthening organisational resilience, innovation, and entrepreneurship in the post-COVID-19 era (Portuguez Castro & Gómez Zermeño, 2020).

In a similar vein, Donaires and Martinelli (2019) suggest that evolutionary management concerns the adaptability factor, which is something insignificant until an enduring and stable economic condition ceases to exist. The earlier one-sided specialisation developed in times of stability by offering efficiency becomes a problem that can threaten future progress. Only then, adaptability turns out to be the significant attribute that can reinforce organisational survival and development (Ulrich & Probst, 1984). In this conceptual direction, evolutionary management is primarily a paradigm instead of a specific model or method, and, therefore, there

is no one best way to perform this in practice. In this context, Donaires and Martinelli (2019, pp. 82–83) offer four specific operating principles of evolution to be considered in the analysis of evolutionary management:

(I) In complex organisational systems, evolution is not a punctual, one-shot process, but it is an overall and continuous process. (II) Evolution is essentially an improvement process. (III) Evolution happens in self-organising and adaptive systems, through a process of self-organisation and adaptation, and should result in systems with enhanced self-organising and adaptive capabilities. (IV) In the case of social systems, evolution must be oriented to higher-level aspects of improvement, including moral and ethical improvement.

Concerning literature that deals, either implicitly or explicitly, with 'the post-COVID-19 management era,' various scholars embrace similar research orientations. Liguori and Pittz (2020), who analyse new strategies for small businesses to survive and thrive in this new era, argue that managing today's disruption needs cost-efficient and aggressive business moves. Verma and Gustafsson (2020) investigate recent emerging research trends in business and management and turn their attention to the need for companies to modify their existing supply chain measures to ensure business continuity while dealing with the COVID-19 crisis. Dessouky and Al-Ghareeb (2020) also observe that resilient organisations can better support their human resources and cope with this newly altered work environment. Finally, in a recent article by Nonaka and Takeuchi (2021), humane business strategy and knowledge management elements are suggested. The authors argue that capitalism as a global socioeconomic system must evolve into a new form of 'wide capitalism' based on phronesis. This 'phronesis' refers to the sought 'new normal,' which means living in harmony with society and nature, pursuing the common good as a way of life and, therefore, in terms of business.

Therefore, it seems that various theoretical approaches exist that can be useful for refocusing today's management theory and practice (Akpan et al., 2020; Carnevale & Hatak, 2020). Contemporary management approaches must increasingly pay interest and attention to co-adjustment with new strategic and technological issues that arise these days, facilitating the processes of rapid and proactive innovation and setting up effective change management mechanisms to deal with the 'ubiquitous' crisis (Kalina, 2020; Motzer et al., 2020). The upcoming concluding section deals with innovation and the corresponding change management mechanisms needed for effective organisational adaptation and survival, counter-proposing the idiosyncratic synthesis of strategy, technology, and management (the Stra.Tech.Man approach).

4. Counterproposal: Change management and innovation in 'Stra.Tech.Man' terms (strategy-technology-management synthesis)

This chapter highlighted the complexity that is increasingly diffused within the theoretical evolution of management. An emphasis was placed on the conceptual foundations laid by classical management scholars, on the first counter-proposed theories of the behavioural approach or human relations, and on the later Japanese philosophy focused on total quality. The current global crisis and restructuring seem to be creating the need for a profoundly renovated management paradigm. By analysing various definitions in this context and focusing on the evolutionary management conception in the post-COVID-19 era, we concluded that today's critical issue is adaptability (via an integrated change management framework) and survival (via a continuous flow of innovation creation and diffusion at all levels). This concluding section suggests a refocused change management and adaptation concept for all organisational levels. This is an issue of increasing signif-

icance, attracting interest from strategic management and innovation and development theorists (Abu El-Ella et al., 2015; Bogers et al., 2019).

Since every organisation uniquely implements management functions, any linear, mechanistic, and self-centric internal process is theoretically shallow and misplaced (Burns & Stalker, 2011; McNamara, 2009). It could be said that effective management means synthesising various elements from an idiosyncratically evolutionary way; therefore, it is far from being an 'endoscopic routine,' applied everywhere and always with the same content and rationale (Pfeffer, 1995). Effective management is never based on easy recipes and is not subject to vague generalisations and one-sided perspectives (Graupp et al., 2020; Kotter, 1977). It emerges as an organic, delicate, and evolutionary process of human, technical, social, and economic dimensions within social entities (private enterprises or other socioeconomic organisations). Effective management is like a 'business medicine,' meaning that every 'patient' business has its physiological specificity, and, therefore, its treatment should be of a corresponding unique character, examining the 'patient' in an integrated way (Vlados & Chatzinikolaou, 2019; Zeleny, 2010).

In this respect, effective management entails complex decision-making, far from any simple mechanistic process of imposing 'identical orders,' regardless of the type of organisation and its specific, dynamically evolving external environment. Effective management must overcome individual differences and contrasts, constantly synthesising towards a higher quality level. Effective management is a process of continuous and dialectic synthesis and re-synthesis within all socioeconomic organisations, aiming to adapt effectively to their environment's constant changes (Langley & Sloan, 2011; Vlados et al., 2019). Therefore, it is noted that contemporary management conceives (or ought to conceive) how organisations coordinate the management functions in their internal environment to innovate as effectively as possible. Exploiting an evolutionary perspective is critical in this context for organisations to understand how the coevolving internal and external organisational environments interact, adapt, and re-adapt in the globalised innovation dynamics.

Management is primarily an internal, reflective, and adaptational business environment process, which, depending on whether it is sophisticated and integrated, can lead to the creation and diffusion of innovation. Innovation is the ultimate goal in every organisation, and, according to a Schumpeterian and neo-Schumpeterian perspective, should be conceptualised as leading to revolutionary changes in all participating socioeconomic systems, at all levels—individual, local, national, international (R. Nelson et al., 2018; Schumpeter, 1942). Contemporary management theory understands the coevolution between the internal and external business environment, based on this system of the five fundamental functions of planning, organising, staffing, control, and coordinating, and the interrelationship between this integrated management with strategy and technology (Figure 3).

[Figure 3 about here]

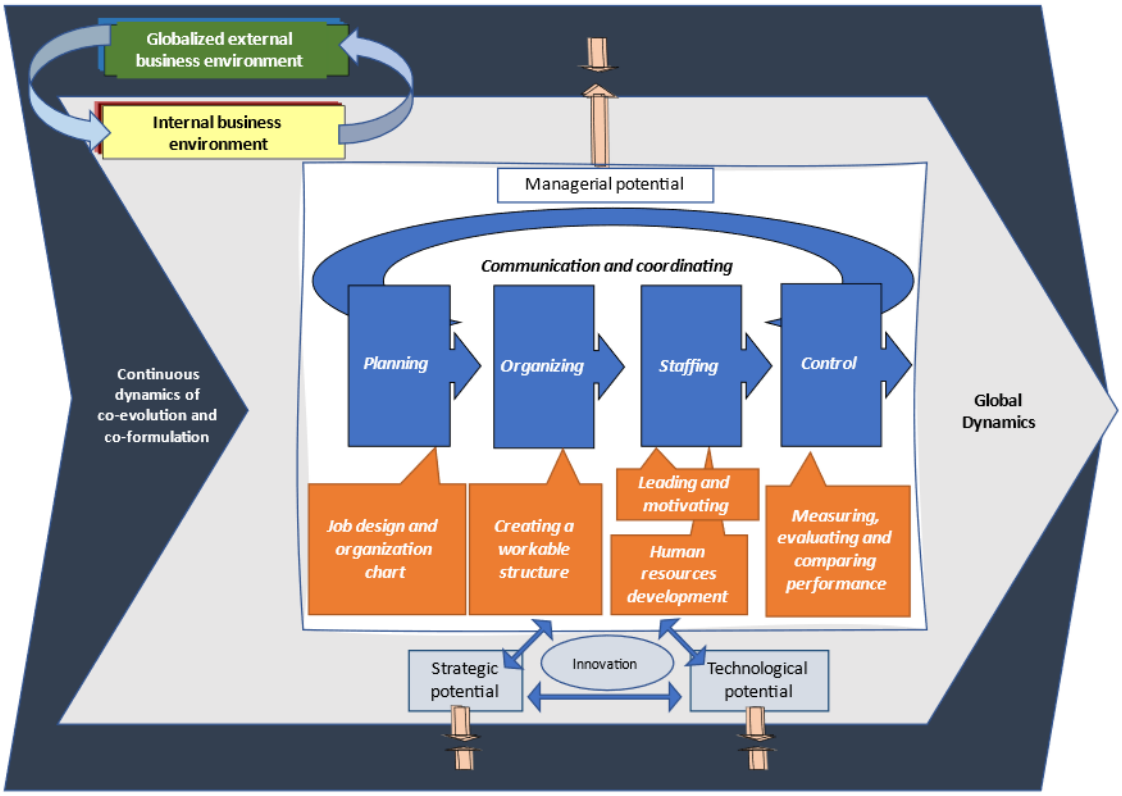


Figure 3. Fundamental functions of contemporary management and the external global environment

The integrated managerial potential first refers to job designing and building a corresponding organisation chart based on specific planning mechanisms. Second, it is about creating a workable structure in terms of organising. Third, it raises the issue of leading, motivating, and developing human resources. Fourth, it corresponds to measuring, evaluating, and comparing performance inside the controlling structures the organisation activates. Fifth, it is about undertaking the right actions to coordinate the entire process. In this evolutionary scheme, innovation creation requires a synthesis of the managerial potential with the related strategic and technological potential because innovation dictates how adaptable an organisation is to the external global dynamics.

However, innovation brings forth explicit or implicit changes inevitably. The question arising from this new integrated strategy-technology-management conceptualisation is how to manage change in an increasingly fluid and uncertain environment like the one that emerges nowadays in the post-COVID-19 era. The handling of change, already from the first relevant approaches (Bennis, 1966), results from preparing the organisational climate, clarifying the need for change, and drawing up and implementing concrete projects. Change management refers to any organisational transformation by using specific methods to redefine the available resources and capabilities (By et al., 2011). Organisational change management understands the organisation in unity and, to this end, analyses and composes the different deliberate changes (Burnes, 2009).

In the 'Stra.Tech.Man' approach, change means transforming a socioeconomic organisation's behaviour to its organic structure and 'physiology.' In this sense, the organisation's 'homeostasis' changes the way it

composes the dimensions of strategy, technology, and management. As suggested by Vlados (2019a, 2019b, 2021), the 'Stra.Tech.Man' theoretical scheme is mostly a business consulting tool aimed at diagnosing the firm's 'physiology,' in the sense that each socioeconomic organisation has specific environmental boundaries, determined by the idiosyncratic ways it effectively synthesises (or not) the coevolving spheres of its strategy, technology, and management. There are three sets of questions that correspond to the combined interpretive perspective of the Stra.Tech.Man approach to innovation:

(1) Strategy: Where is the organisation? Where does it desire to go? How does it go there? Why?

(2) Technology: How does the organisation draw, create, compose, diffuse, and reproduce its expertise and knowledge? Why?

(3) Management: How does the organisation use its available resources? Why?

From the 'Stra.Tech.Man' perspective, the firm's physiological evolution and 'mutation' is the analytical epicentre of the entire innovation and development process; this is the motor of continuous transformation for both the firm and the surrounding socioeconomic environment. However, innovation requires change management mechanisms to be adequately diffused in socioeconomic organisations and systems. Specific phases of change management have been distinguished in the 'Stra.Tech.Man approach,' considered critical to innovation since they constitute a generalisation of the firm's principles. Stra.Tech.Man change management consists of five distinct and successive phases, each with eight steps, defining a continuously elevating and ongoing process for the firm. In this conceptual scheme that can be used for business consulting in today's organisations, the steps and suggestions for the process of managing organisational change are as follows:

- Strategic mutation (Hill & Jones, 2004; Yaeger & Sorensen, 2009): First, clarify and deepen the vision and mission developed by the firm. Second, doubt about any strategic certainties and inform all stakeholders. Third, build mechanisms to understand how the external environment changes as early and comprehensively as possible. Fourth, develop an understanding of the firm's internal environment. Fifth, build a correlative and evolutionary SWOT analysis. Sixth, carefully build any alternatives and evaluate them. Seventh, choose the best-suited (both ambitious and realistic) strategy. Eighth, comprehensively and cohesively analyse the organisation's tactics and politics.

- Technological mutation (Abbott, 2013; Abdelnour-Nocera, 2013): First, profoundly understand the technological rationale implemented in the organisation. Second, gain a comparative picture of the organisation's technological capabilities. Third, develop technology learning processes, and fourth, stimulate the organisation's mechanisms for creating technological knowledge. Fifth, reinforce the processes for technology diffusion. Sixth, practically support the implementation of modern technology. Seventh, do not be afraid of experimentation if it leads to 'substantial lessons.' Eighth, control, evaluate and, if possible, reward the successful implementation of contemporary technology.

- Management mutation (Carmichael, 2011; Senge, 1990; Tobin & Pettingell, 2008): First, experiment with new planning methods. Second, make the organisation chart lighter. Third, build a meritocratic way of placing the right person at the right time in the right place. Fourth, give the organisation's people the leaders who suit them and inspire them. Fifth, make the organisation a 'learning organisation.' Sixth, give better, specialised, and rewarding motivation. Seventh, measure and evaluate in a com-

parative and fair spirit. Eighth, open new communication channels and build new ways to coordinate action.

- Innovative synthesis (Fonseca, 2002; Szirmai et al., 2011): First, clarify the achieved transformations in strategy, technology and management and cautiously prepare the new Stra.Tech.Man synthesis. Second, attribute weight, balance, and adjust the Stra.Tech.Man triangle of innovation. Third, spread the 'revolution' message and build a dynamic leadership team. Fourth, remove obstacles, commission roles, and encourage. Fifth, maintain balance in implementing the 'intervention.' Sixth, try getting 'easy' wins and not over-celebrate them. Seventh, set up control and evaluation points for the entire effort. Eighth, in the end, do not forget to reward those who fought for this change.
- Assimilation of change (Hayes, 2018; Raina, 2018): First, preserve the acts that brought results and unify them. Second, do not punish those who experimented honestly but failed but those who did not fight at all. Third, refresh the hierarchy, bringing in new people. Fourth, avoid conservatism by making 'yesterday's success' a goal to overcome. Fifth, bring 'foreigners' into the organisation and withstand their criticisms. Sixth, build an organisation that can be 'loved.' Seventh, do not rest on the laurels of earlier success. Eighth, always start all over again (Figure 4).

[Figure 4 about here]

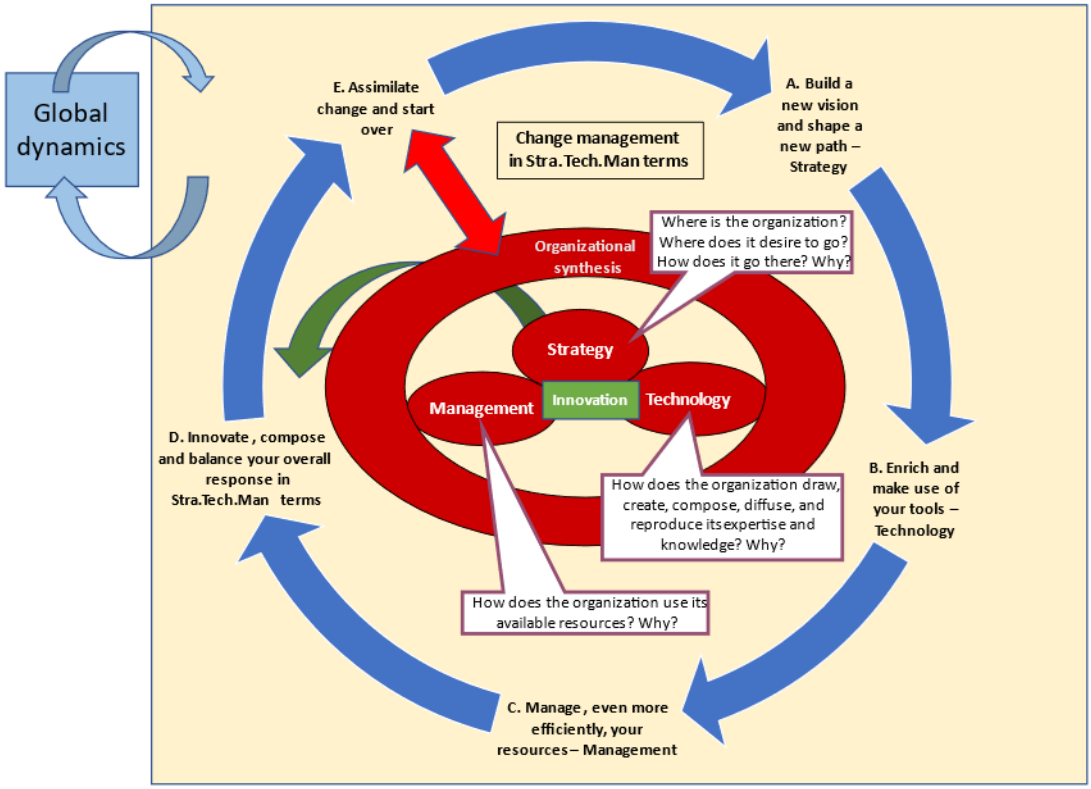


Figure 4: The Stra.Tech.Man change management. Based on Vlados (2019c)

These steps are suggested as instruments for effective change management within any organisation called

upon to adapt to global dynamics, exploiting the specific survival potential (innovation) that the organisational synthesis of strategy, technology, management can offer. This methodology could be applied and exploited within different organisational mechanisms, possibly of different sizes and sectoral focus. This Stra.Tech.Man application for managing change and innovation within various organisations could be an avenue for future research via action research methodologies or related self-evaluation or external evaluation business research techniques (Antwi & Hamza, 2015; Krivokapic-Skoko & O'Neill, 2011).

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